JA

Joshua Alcobia Gomes

Mechanical Engineer

Second-year MEng Mechanical Engineering student at UCL (predicted First), heading to McLaren on an industrial placement for 2026–27. I work across automotive and aerospace — from CFD and rocketry to pharmaceutical modelling, machine learning and AI tooling.

DISCIPLINE
Mechanical Engineering

UNIVERSITY
UCL · MEng

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ABOUT ME

Curiosity, engineered.

I'm a mechanical engineer drawn to the fastest, most demanding corners of the field — motorsport and flight. I'm in my second year of an MEng in Mechanical Engineering at UCL (predicted First, 83%), and I'll be joining McLaren on an industrial placement for the 2026–27 academic year.

My work spans design, simulation, testing and the reality of making things perform: CFD in ANSYS Fluent, structural and generative design in Fusion 360 and SolidWorks, and machine-learning and AI tooling built during my scientific-modelling internship at Novartis. I lead teams as happily as I run the numbers — from an 8-engineer national rocketry squad to award-winning design challenges.

Away from engineering I compete nationally in track and field, which is where a lot of the discipline behind this work comes from — and the calm, natural palette of this site reflects where I recharge.

Fusion 360 · SolidWorks · Onshape | ANSYS Fluent & Mechanical | COMSOL · OpenRocket · QBlade | Python · MATLAB · C++ | CFD & FEA | Machine Learning · LLMs | 3D printing · laser cutting

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AT A GLANCE

Discipline: MEng Mechanical Engineering
University: UCL · 2nd year (predicted 1st, 83%)
Next: McLaren — Industrial Placement 2026–27
Focus: Automotive & Aerospace
Based in: London, UK
Email: jg7alcobia@gmail.com

CONTENTS

A separate 2-page résumé accompanies this portfolio.

01 · AUTOMOTIVE / MOTORSPORT

McLaren

High-performance automotive engineering — my industrial placement year.

ROLE	TEAM	PERIOD	DOMAIN
Industrial Placement	**TBC**	**2026–27 academic year**	**Automotive · Motorsport**

I'll be joining McLaren for my industrial placement across the 2026–27 academic year — a full year embedded in one of the sharpest engineering environments in the world. This page will grow into a record of the projects I work on there.

APPROACH

1. Bring rigorous simulation and design fundamentals from UCL and NING Research.
2. Learn the pace and precision of elite motorsport engineering.
3. Document each project here as the placement progresses.

PROJECTS

3 projects

COMING SOON

Placement project 1

Project details will appear here once my placement begins in 2026.

COMING SOON

Placement project 2

Project details will appear here once my placement begins in 2026.

COMING SOON

Placement project 3

Project details will appear here once my placement begins in 2026.

02 · RESEARCH & DEVELOPMENT

NING Research

Applied CFD and engineering simulation — turning real client problems into validated models.

ROLE	FIELD	PERIOD	DOMAIN
Simulation / CFD Engineer	**CFD & Engineering Simulation**	**Start – present**	**R&D · Consulting**

At NING Research I work on computational fluid dynamics and engineering-simulation projects for a range of clients — from ventilation and mixing to structural and environmental analysis.

APPROACH

1. Define the physical problem and success criteria with the client.
2. Build and mesh the CFD / simulation model (OpenFOAM, ANSYS).
3. Validate against data, then translate results into clear recommendations.

PROJECTS

12 projects

CFD · VENTILATION

Spacelogic Coughing Simulation

Add a short description of this project and its outcome.

CFD · MIXING

OpenFOAM Mixing Tank

Add a short description of this project and its outcome.

SIMULATION

Topayoh

Add a short description of this project and its outcome.

CFD · BUILT ENVIRONMENT

80 Kim Yam Road

Add a short description of this project and its outcome.

CFD · ENVIRONMENTAL

Roof Rainfall Project

Add a short description of this project and its outcome.

CFD · MIXING

Sanli Mixing Tank

Add a short description of this project and its outcome.

SIMULATION · SAFETY

Safe Site — 3logytech

Add a short description of this project and its outcome.

CFD · SPORTS AERO

KiteFoil — Max Maeder (SportSG)

Add a short description of this project and its outcome.

SIMULATION

Project 9

Add this project's title and description.

PROCESS SIMULATION

Pad Production Line (Kenvue)

Add a short description of this project and its outcome.

SIMULATION

Project 11

Add this project's title and description.

SIMULATION

Project 12

Add this project's title and description.

03 · PHARMA / MACHINE LEARNING

Novartis

Scientific Modelling Intern — machine learning, simulation and AI tooling in pharmaceutical manufacturing.

ROLE	PERIOD	LOCATION	DOMAIN
Scientific Modelling Intern	**Aug – Sept 2025**	**Basel, Switzerland**	**R&D · Pharma / ML**

Over a summer internship in Basel I built machine-learning models, simulations and AI tools that removed manual work and cut costs across pharmaceutical R&D — spanning powder characterisation, process analytics and LLM-based automation.

APPROACH

1. Turn messy sensor and process data into clean, usable models.
2. Automate repetitive analysis with Python, dashboards and LLM workflows.
3. Validate against real batches and quantify the time and cost saved.

PROJECTS

5 projects

MACHINE LEARNING

ML Powder Flow Classification

Machine-learning models to predict powder flow characteristics from experimental data.

- Used LazyClassifier for rapid model selection, then Grid Search, Lasso regularisation and Bayesian optimisation for hyperparameter tuning.
- Saved CHF 40,000 per batch of APIs by predicting behaviour ahead of manufacture.

AI INFRASTRUCTURE

MCP Servers for Internal AI Tooling

Add a short description of the MCP server work — what it connected and enabled.

DATA / PYTHON

Streamlit Analytics Dashboards

Interactive Streamlit applications that eliminated manual data searching for the department.

- Built dropdown-driven apps for dwell-time analysis and screw customisation.
- Developed Python programs for Loss-on-Drying calculations from sensor data, solving mass-balance equations for moisture content and drying efficiency.

LLMS · AGENTS

AI Project Summariser (LLMs + Agentic Workflows)

An AI summariser that gathers data from internal and external databases automatically.

- Used automated workflows and LLMs to compile project context on demand.
- Reduced research time from ~50 minutes to ~15, saving weeks of cumulative effort.

CAD · SIMULATION

Pill CAD Design for FEA / COMSOL

Tablet geometry modelled for high-performance-compute simulation.

- Designed tablets in SolidWorks from technical drawings for COMSOL HPC simulation.
- Saved thousands per batch through predictive modelling.

04 · EDUCATION

UCL

MEng Mechanical Engineering — the competition teams and projects that shaped how I engineer.

DEGREE	YEARS	PREDICTED	DOMAIN
MEng Mechanical Engineering	**Sept 2024 – June 2028**	**First Class (83%)**	**Education**

I'm in my second year of an MEng in Mechanical Engineering at UCL, predicted a First (83%). Alongside the degree I throw myself into engineering teams and competitions — rocketry, motorsport, design challenges and more — where the theory meets real hardware.

APPROACH

1. Apply core mechanical engineering: thermofluids, solid mechanics, dynamics, materials.
2. Specialise toward automotive and aerospace through team projects.
3. Lead and deliver — from budgets and Gantt charts to manufacture and test.

PROJECTS

7 projects

AEROSPACE · TEAM LEAD

UCL Racing — Rocket

National Rocketry Championship team lead, aerostructures & simulations engineer.

- Led an 8-engineer team with a £1,000 budget to 8th of 50 at the National Rocketry Championship.
- Validated ANSYS Fluent CFD against drag-coefficient data to 97% accuracy.
- Designed the aerostructure in Fusion 360, increasing apogee by 203 m.
- Ran the project in Jira with Gantt charts, risk assessments and sponsorship outreach.

AUTOMOTIVE · MOTORSPORT

Formula Student

Add your Formula Student role, subsystem and contributions.

DESIGN & MANUFACTURE

IMechE Design Challenge

Design & manufacturing engineer — runner-up, Greater London Regional Final.

- Secured National Final qualification as one of only two teams to achieve perfect accuracy.
- Won both the Design Award and the Challenge Competition.
- Used Fusion 360 Generative Design to cut mass by 115.87 g (60%) at a safety factor of 2.
- Produced a full Bill of Materials with data visualisation to identify cost drivers.

TURBINES · COMPETITION

UCL Mechathon 2025

Won the competition (team of 6) with a 76 mm-radius wind-turbine design.

- Achieved 29% efficiency at 3.3 V.
- Used QBlade to analyse aerofoil lift-to-drag ratios and optimal angle of attack (95.8% accurate).
- Developed manufacturing methods across milling, 3D printing and laser cutting.

COURSEWORK

Project 5

Add another UCL project — title, one line, and a few bullets.

COURSEWORK

Project 6

Add another UCL project — title, one line, and a few bullets.

COURSEWORK

Project 7

Add another UCL project — title, one line, and a few bullets.

PRESENTATIONS & COURSEWORK

Drop presentation PowerPoints and coursework PDFs into assets/docs/ and list them here.

Presentation / coursework 1

Presentation / coursework 2

05 · PERSONAL BUILDS

Side Projects

The things I build for the love of it — research, robotics and everything between.

THEMES
Aerospace · Robotics · Research

TOOLS
CAD · Arduino · C++ · 3D print

STATUS
Ongoing

A selection of personal and academic projects I've taken end-to-end — from theoretical research papers to fully built, programmed hardware.

APPROACH

1. Start from a question or a problem worth solving.
2. Model, build and iterate — CAD, electronics, code and test.
3. Document the result properly, challenges and all.

PROJECTS

4 projects

RESEARCH · IMPACT

Internal Pressure of Thin-Walled Vessels During Collision

Theoretical research and mathematical modelling on pressure vs. impact force.

- Modelled the relationship between internal pressure and impact force.
- Wrote a 4,000-word research paper on optimal pressure for impact reduction and its applications to safety systems.

ROBOTICS · AUTOMATION

Automated Fruit & Vegetable Cutting Robot

A kitchen robot designed, built and programmed from scratch.

- Built and programmed the robot with Arduino and C++.
- Modelled and optimised components in CAD for functionality and durability.
- Assembled and iteratively refined the mechanical systems and control algorithms.

MAKING

Side project 3

Add a personal build — title, one line, and a few bullets.

MAKING

Side project 4

Add a personal build — title, one line, and a few bullets.